Convex Sets, Convex Functions, and Hyperplanes

A study guide for the material of Homework 1

Optimization (Caramanis & Sanghavi) — companion notes

September 6, 2026

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1 Why convexity is the whole game

An optimization problem $\min_{x \in \mathcal{F}} f(x)$ is *convex* when the feasible set $\mathcal{F}$ is a convex set and the objective f is a convex function. Two facts make this class special:

1. **Local = global.** Every local minimizer of a convex problem is a global minimizer. Descent methods therefore cannot get "stuck."

2. **Certifiable optimality.** There is a clean first-order optimality condition ($\nabla f(x^\star)^\mathsf{T}(y-x^\star) \geq 0$ for all feasible y) and a duality theory that produces *proofs* of optimality.

Everything in this unit is machinery for recognising, building, and manipulating convex sets and functions. The homework tests three skills that recur all semester:

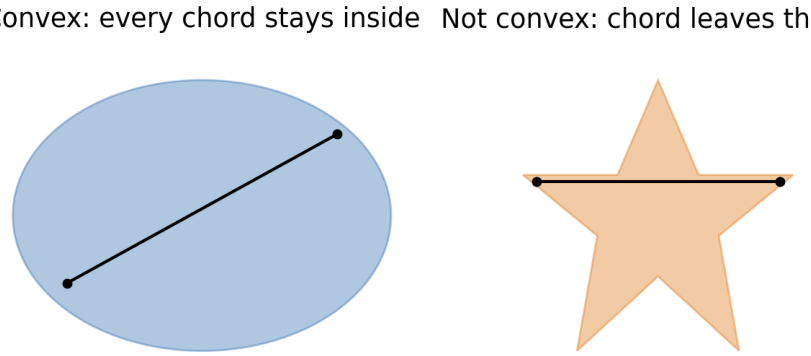
- proving a set or function is convex *from the definition*;
- knowing which *operations preserve* convexity (and which do not);
- translating geometric statements ("closer to v_1 than v_2") into algebraic ones ("$c^\mathsf{T}x \leq d$").

2 Convex sets

2.1 Definitions

Definition 2.1 (Convex set)**.** $C \subseteq \mathbb{R}^n$ is *convex* if for all $x, y \in C$ and all $\lambda \in [0,1]$, $\lambda x+(1-\lambda)y \in C$.

The point $\lambda x + (1-\lambda)y$ traces the closed segment from y ($\lambda = 0$) to x ($\lambda = 1$). So "convex" means: *the segment joining any two points of the set stays in the set.* The empty set, a single point, and all of $\mathbb{R}^n$ are convex.



Definition 2.2 (Related notions)**.**

- A *convex combination* of $x_1, \ldots, x_k$ is $\sum_i \theta_i x_i$ with $\theta_i \geq 0$, $\sum_i \theta_i = 1$. A set is convex iff it is closed under all convex combinations (induction on k).
- The *convex hull* $\operatorname{conv} S$ is the set of all convex combinations of points of S; it is the smallest convex set containing S.
- C is *affine* if it contains the whole *line* through any two of its points ($\lambda \in \mathbb{R}$). Affine $\Rightarrow$ convex.
- C is a *cone* if $x \in C,\ \theta \geq 0 \Rightarrow \theta x \in C$; a *convex cone* is closed under all conic combinations $\theta_1 x_1 + \theta_2 x_2$, $\theta_i \geq 0$.

2.2 The examples you must know

Set	Formula	Why convex
Hyperplane	$\{x : a^\mathsf{T} x = b\}$, $a \neq 0$	affine
Half-space	$\{x : a^\mathsf{T} x \leq b\}$, $a \neq 0$	linear inequality survives averaging
Euclidean ball	$\{x : \|x - x_c\|_2 \leq r\}$	triangle inequality
Norm ball	$\{x : \|x - x_c\| \leq r\}$, any norm	triangle inequality
Ellipsoid	$\{x : (x - x_c)^\mathsf{T} P^{-1}(x - x_c) \leq 1\}$, $P \succ 0$	image of ball under $P^{1/2}$
Norm cone	$\{(x, t) : \|x\| \leq t\}$	sublevel set of convex $\|x\| - t$
Polyhedron	$\{x : Ax \preceq b,\ Cx = d\}$	intersection of half-spaces/hyperplanes
Simplex	$\mathrm{conv}\{v_0, \ldots, v_k\}$	convex hull
PSD cone $\mathbb{S}^n_+$	$\{X = X^\mathsf{T} : X \succeq 0\}$	$z^\mathsf{T}(\lambda X + (1-\lambda)Y)z \geq 0$
Voronoi cell of v_1	$\{x : \|x - v_1\| \leq \|x - v_j\|\ \forall j\}$	intersection of half-spaces (HW1 P3)

2.3 Operations that preserve convexity of sets

Theorem 2.3 (Calculus of convex sets)**.**

1. ***Intersection.*** *If C_α is convex for every α in an arbitrary index set, then $\bigcap_\alpha C_\alpha$ is convex.* (HW1 P1a.)

2. ***Affine image and preimage.*** *If $f(x) = Ax + b$ is affine, C convex $\Rightarrow f(C)$ convex, and D convex $\Rightarrow f^{-1}(D) = \{x : Ax + b \in D\}$ convex. Special cases: scaling, translation, projection onto coordinates, Minkowski sum $C_1 + C_2$, Cartesian product $C_1 \times C_2$.*

3. ***Perspective and linear-fractional maps*** *also preserve convexity (B&V §2.3.3); rarely tested at this stage.*

Proof of 1. Take $x, y \in \bigcap_\alpha C_\alpha$, $\lambda \in [0, 1]$. For each α, $x, y \in C_\alpha$ and C_α convex give $\lambda x + (1-\lambda)y \in C_\alpha$. True for every α, hence true for the intersection. □

Proof of 2 (preimage). If $Ax + b \in D$ and $Ay + b \in D$, then $A(\lambda x + (1-\lambda)y) + b = \lambda(Ax + b) + (1-\lambda)(Ay + b) \in D$ by convexity of D. The image case is similar. □

Common pitfall. Unions and complements do not preserve convexity. $[0, 1] \cup [2, 3]$ is not convex; the complement of a disk is not convex. When you see "union" or "or" in a set description, expect non-convexity unless the pieces overlap heavily. *(HW1 P1b.)*

Key idea. To prove a complicated set is convex, *do not* start from the definition. Instead, write it as (i) an intersection of simpler convex sets, or (ii) an affine preimage of a standard convex set, or (iii) a sublevel set of a convex function. Example: $\{x : \|Ax + b\|_2 \leq c^\mathsf{T} x + d\}$ is the preimage of the second-order cone under the affine map $x \mapsto (Ax + b,\ c^\mathsf{T} x + d)$. *(HW1 P4.)*

3 Hyperplanes and half-spaces

3.1 Geometry of $a^\mathsf{T}x = b$

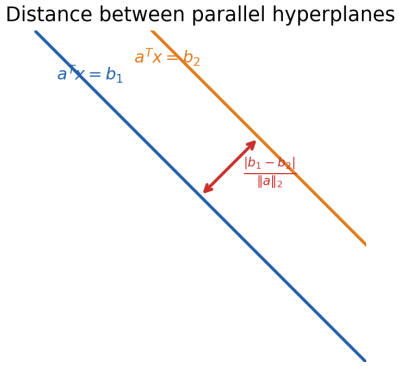
Fix $a \neq 0$. The hyperplane $H = \{x : a^\mathsf{T}x = b\}$ is the set of x with a fixed inner product with a. Equivalently, picking any $x_0 \in H$, $H = \{x : a^\mathsf{T}(x - x_0) = 0\} = x_0 + a^\perp$: it is the translate of the $(n-1)$-dimensional subspace orthogonal to a. The vector a is the *normal*; it points into the half-space $\{a^\mathsf{T}x \geq b\}$ and away from $\{a^\mathsf{T}x \leq b\}$.

Proposition 3.1 (Distance formulas)**.**

1. *Distance from a point x to $H = \{a^\mathsf{T}x = b\}$:* $\operatorname{dist}(x, H) = \dfrac{|a^\mathsf{T}x - b|}{\|a\|_2}$. *The closest point is* $x - \dfrac{a^\mathsf{T}x - b}{\|a\|^2}a$.

2. *Distance between parallel hyperplanes $\{a^\mathsf{T}x = b_1\}$, $\{a^\mathsf{T}x = b_2\}$:* $\dfrac{|b_1 - b_2|}{\|a\|_2}$. (HW1 P5b.)

Mnemonic: normalise a to a unit vector; then b is the signed offset of the plane along a.



3.2 Containment and equality of half-spaces

Proposition 3.2 (B&V 2.6, HW1 P5a)**.** *For $a, \tilde{a} \neq 0$:*

$$\{a^\mathsf{T}x \leq b\} \subseteq \{\tilde{a}^\mathsf{T}x \leq \tilde{b}\} \iff \tilde{a} = \lambda a \text{ and } \tilde{b} \geq \lambda b \text{ for some } \lambda > 0,$$

with equality of the half-spaces iff additionally $\tilde{b} = \lambda b$.

The proof strategy is instructive: assume containment, then *walk to infinity* inside the small half-space along a direction $w \perp a$ (to show $\tilde{a} \parallel a$) and along $-a$ (to show the multiple is positive). Any direction in which the small half-space is unbounded must also leave the big one unbounded.

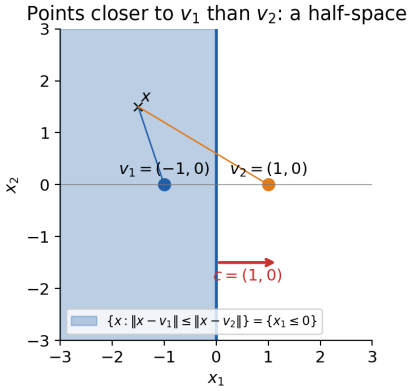
3.3 Half-spaces from distance comparisons

Proposition 3.3 (Perpendicular bisector, HW1 P3)**.** *For $v_1 \neq v_2 \in \mathbb{R}^n$,*

$$\{x : \|x - v_1\|_2 \leq \|x - v_2\|_2\} = \{x : c^\mathsf{T}x \leq d\}, \qquad c = 2(v_2 - v_1), \quad d = \|v_2\|^2 - \|v_1\|^2.$$

Idea. Square both sides (allowed: both are ≥ 0), expand $\|x - v\|^2 = \|x\|^2 - 2v^\mathsf{T}x + \|v\|^2$, and watch the quadratic term $\|x\|^2$ cancel. What remains is linear in x. □

The boundary passes through the midpoint $\frac{1}{2}(v_1+v_2)$ with normal v_2-v_1: it is the perpendicular bisector. Intersecting over many v_j gives the Voronoi cell, convex by Theorem 2.3(1).



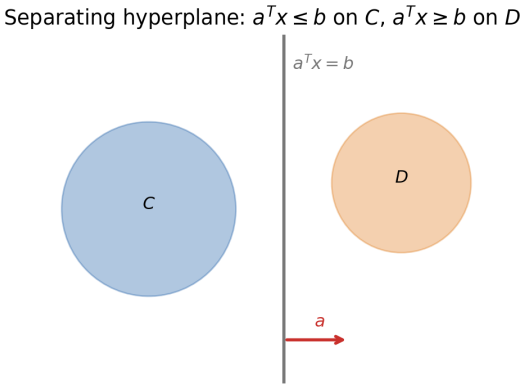
Common pitfall. Sign and direction of c. The normal $c = 2(v_2 - v_1)$ points from v_1 *toward* v_2, and the inequality is $c^{\mathsf{T}}x \le d$. It is easy to flip a sign; always sanity-check with $x = v_1$ (must satisfy the inequality) and $x = v_2$ (must violate it unless $v_1 = v_2$). Also, c and d are only defined up to a common positive factor.

4 Separating and supporting hyperplanes

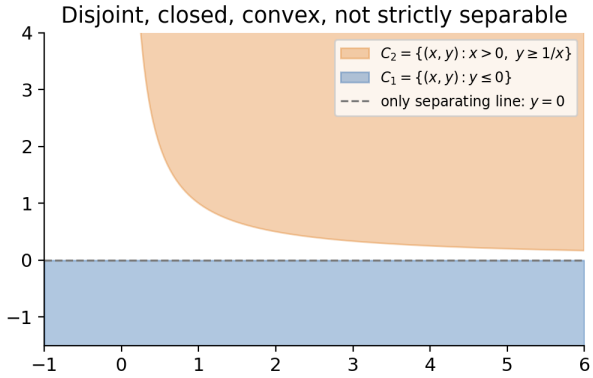
Definition 4.1. Hyperplane $\{a^{\mathsf{T}}x = b\}$ $(a \neq 0)$ *separates* C and D if $a^{\mathsf{T}}x \le b$ on C and $a^{\mathsf{T}}x \ge b$ on D. It *strictly separates* them if both inequalities are strict.

Theorem 4.2 (Separating hyperplane theorem)**.** *If $C, D \subseteq \mathbb{R}^n$ are nonempty, convex, and disjoint, there exists a separating hyperplane.*

Theorem 4.3 (Strict separation)**.** *If in addition C is **closed** and D is **compact** (closed and bounded), the separation can be taken strict; in fact one can separate them by a slab of positive width.*



Common pitfall. Closedness of both sets is *not* enough for strict separation. The standard counterexample (HW1 P2b, B&V 2.23): $C_1 = \{y \le 0\}$ and $C_2 = \{x > 0,\ y \ge 1/x\}$ in $\mathbb{R}^2$ are closed, convex, disjoint, but $\operatorname{dist}(C_1, C_2) = 0$ because $(k, 0) \in C_1$ and $(k, 1/k) \in C_2$ get arbitrarily close. Only the non-strict separator $y = 0$ exists. The underlying reason: two closed sets can have zero distance without intersecting when neither is bounded.



Theorem 4.4 (Supporting hyperplane theorem)**.** *If C is convex and x_0 is a boundary point of C, there is $a \neq 0$ with $a^\mathsf{T} x \le a^\mathsf{T} x_0$ for all $x \in C$. Conversely, a closed set with nonempty interior that has a supporting hyperplane at every boundary point is convex.*

Proof sketch of the strict version, useful to remember: minimise $\|c - d\|$ over $c \in C, d \in D$ (a minimiser exists by compactness), and take the hyperplane through the midpoint of the minimising pair, normal $d - c$. The obstruction in the counterexample above is exactly that this minimum is not attained.

5 Convex functions

5.1 Definition and the epigraph

Definition 5.1 (Convex function)**.** $f : \mathbb{R}^n \to \mathbb{R}$ (more generally, f with convex domain $\operatorname{dom} f$) is *convex* if for all $x, y \in \operatorname{dom} f$ and $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \le \lambda f(x) + (1 - \lambda) f(y).$$

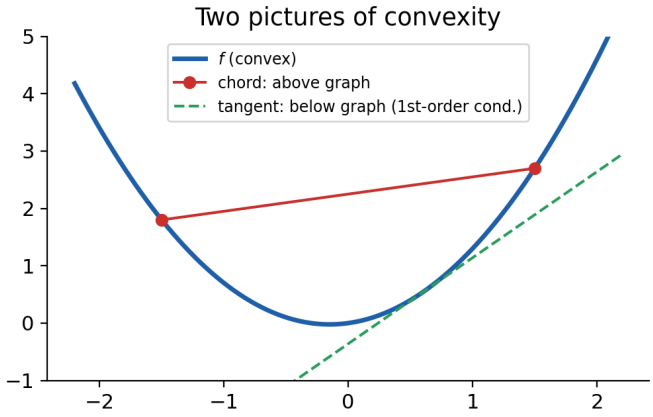
It is *strictly convex* if the inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$; *concave* if $-f$ is convex.

Geometrically: the chord between any two points on the graph lies *on or above* the graph.

Definition 5.2 (Epigraph and sublevel sets)**.** $\operatorname{epi} f = \{(x, t) \in \mathbb{R}^{n+1} : t \ge f(x)\}$ is the region above the graph. The c-sublevel set is $L_c = \{x : f(x) \le c\}$.

Theorem 5.3. *f is convex $\iff$ $\operatorname{epi} f$ is a convex set.*

This is the bridge between the "sets" and "functions" halves of the theory: every set result becomes a function result by applying it to epigraphs.



5.2 Tests for convexity

Theorem 5.4 (Restriction to lines)**.** *f is convex iff for every $x \in \operatorname{dom} f$ and direction v, the univariate function $g(t) = f(x + tv)$ is convex on $\{t : x + tv \in \operatorname{dom} f\}$.*

Theorem 5.5 (First-order condition)**.** *If f is differentiable with convex domain, f is convex iff $f(y) \geq f(x) + \nabla f(x)^{\mathsf{T}}(y - x)$ for all x, y. (Tangent planes lie below the graph; the linearisation is a* global *underestimator.)*

Theorem 5.6 (Second-order condition)**.** *If f is twice differentiable with convex domain, f is convex iff $\nabla^2 f(x) \succeq 0$ for all $x \in \operatorname{dom} f$. In one variable: $f'' \geq 0$. If $\nabla^2 f \succ 0$ everywhere then f is strictly convex (the converse fails: x^4).*

5.3 Examples

Convex	Concave
e^{ax} $(a \in \mathbb{R})$; x^p on $x > 0$ for $p \geq 1$ or $p \leq 0$; $\lvert x\rvert^p$, $p \geq 1$	$\log x$; x^p on $x > 0$, $0 \leq p \leq 1$
every norm $\lVert x\rVert$; $\max_i x_i$; $\log \sum_i e^{x_i}$	geometric mean $(\prod x_i)^{1/n}$ on $\mathbb{R}^n_{++}$
$x^{\mathsf{T}} P x$ with $P \succeq 0$; $a^{\mathsf{T}} x + b$ (both)	$\log\det X$ on $\mathbb{S}^n_{++}$
x^2/y on $y > 0$; $-\log x$	$\min_i x_i$; entropy $-x \log x$ on $x > 0$

5.4 Operations that preserve convexity of functions

Theorem 5.7 (Calculus of convex functions)**.** *Let $f, f_1, \ldots, f_k$ be convex.*

1. ***Nonnegative weighted sums.*** *$\sum_i w_i f_i$ with $w_i \geq 0$ is convex.*
2. ***Composition with affine map.*** *$g(x) = f(Ax + b)$ is convex.*
3. ***Pointwise maximum / supremum.*** *$\max\{f_1, \ldots, f_k\}$ and $\sup_\alpha f_\alpha$ are convex.* (HW1 P1c.)
4. ***Composition rules.*** *$h \circ g$ is convex if h convex nondecreasing and g convex; or h convex nonincreasing and g concave (with $\tilde{h}$ the extended-value extension).*
5. ***Partial minimisation.*** *If $f(x, y)$ is jointly convex and C convex, $g(x) = \inf_{y \in C} f(x, y)$ is convex.*

6. ***Perspective.*** *$g(x,t) = t\,f(x/t)$ on $t>0$ is convex.*

Proof of 3 (the HW1 argument). Let $z = \lambda x + (1-\lambda)y$. For each i: $f_i(z) \le \lambda f_i(x) + (1-\lambda) f_i(y) \le \lambda f_{\max}(x) + (1-\lambda) f_{\max}(y)$. Take the max over i on the left. Equivalently: $\operatorname{epi} f_{\max} = \bigcap_i \operatorname{epi} f_i$, an intersection of convex sets. □

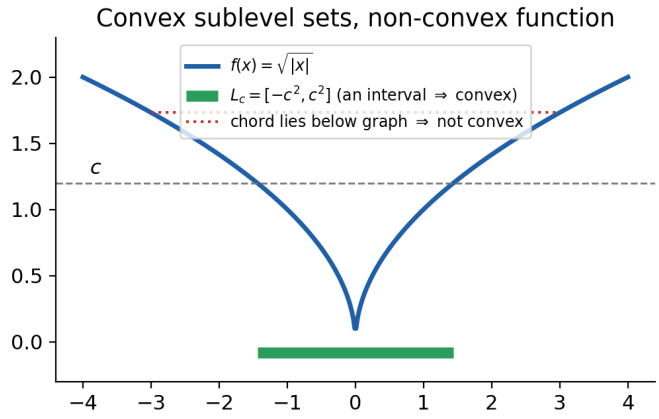
Common pitfall. **Not preserved:** pointwise **minimum** ($\min\{x^2, (x-2)^2\}$ has a bump; $\operatorname{epi} f_{\min} = \bigcup \operatorname{epi} f_i$ is a union), **products** ($x \cdot x = x^2$ fine, but $x \cdot (-x) = -x^2$ is not), **differences** ($f_1 - f_2$ is a difference of convex functions, "DC," generally non-convex), and **arbitrary compositions** (e^{-x} is convex, $-x^2$ concave, but e^{-x^2} is neither). *(HW1 P2a.)*

6 Sublevel sets and quasiconvexity

Proposition 6.1 (HW1 P2c)**.** *If f is convex, every sublevel set $L_c = \{x : f(x) \le c\}$ is convex.*

Proof. $x, y \in L_c \Rightarrow f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda) f(y) \le \lambda c + (1-\lambda)c = c$. □

Common pitfall. **The converse is false.** Convex sublevel sets do *not* imply a convex function. Counterexamples in one variable: $f(x) = \sqrt{|x|}$ (sublevel sets $[-c^2, c^2]$; but $f(2) = \sqrt{2} > 1 = \frac{1}{2}f(0) + \frac{1}{2}f(4)$), $f(x) = x^3$, $f(x) = -e^{-x^2}$, and indeed *any monotone* function. In one variable, "all sublevel sets convex" just means "all sublevel sets are intervals," which is far weaker than convexity.



Definition 6.2 (Quasiconvex)**.** f is *quasiconvex* if every sublevel set is convex; equivalently $f(\lambda x + (1-\lambda)y) \le \max\{f(x), f(y)\}$.

Hierarchy: convex ⇒ quasiconvex, and the implication is strict. Quasiconvex problems can still be solved (by bisection on c plus a convex feasibility problem for each L_c), which is why sublevel sets matter beyond this homework.

Key idea. The right mental picture of "f convex $\Rightarrow$ L_c convex": slice the convex epigraph with the horizontal hyperplane $t = c$ and project down. Slicing (intersection with a convex set) and projecting (affine image) both preserve convexity. The converse fails because the sublevel sets are only *horizontal* slices; convexity of epi f needs every slice, in every direction.

7 Proof toolkit

Proof template. *Set C is convex (from the definition).*

1. "Let $x, y \in C$ and $\lambda \in [0, 1]$. Set $z = \lambda x + (1 - \lambda)y$."
2. Unpack what $x \in C$ and $y \in C$ mean (an inequality, an equation, membership in other sets).
3. Show z satisfies the same condition, using: linearity ($a^\mathsf{T} z = \lambda a^\mathsf{T} x + (1 - \lambda)a^\mathsf{T} y$), the triangle inequality plus homogeneity for norms ($\|\lambda u + (1 - \lambda)v\| \leq \lambda\|u\| + (1 - \lambda)\|v\|$), or convexity of ingredient functions.
4. "Hence $z \in C$, so C is convex."

Proof template. *Function f is convex (from the definition).* Same skeleton; the target inequality is $f(z) \leq \lambda f(x) + (1 - \lambda)f(y)$. Chain inequalities: apply convexity of the pieces first, then bound each piece by something involving $f(x), f(y)$, keeping the weights $\lambda, 1 - \lambda \geq 0$ in front.

Proof template. *Something is NOT convex.* One explicit witness suffices: specific x, y, λ (usually $\lambda = \frac{1}{2}$ and small integers) for which the defining condition fails. Compute both sides numerically and state the strict inequality. Draw the picture to find the witness: look for a "bump" (functions) or a "notch/gap" (sets).

Proof template. *Geometric $\rightarrow$ algebraic ("show this region is a half-space").* Write the geometric condition as an inequality, square if both sides are nonnegative, expand, cancel the nonlinear terms, and read off c and d. Verify with two test points.

Proof template. *No strict separation exists.* Assume $a^\mathsf{T} p < b < a^\mathsf{T} q$ for all $p \in C$, $q \in D$. Send points of C to infinity in various directions to force components of a to vanish or have a sign; then find a sequence in D whose $a^\mathsf{T} q$ tends to a value $\leq b$. Contradiction.

8 Self-test questions

Try these without looking back; short answers follow.

Self-test 8.1. Is $\{x \in \mathbb{R}^n : \|x\|_2 \leq 1\} \cap \{x : x_1 \geq 0\}$ convex? Is $\{x : \|x\|_2 \leq 1\} \cup \{x : \|x - 3e_1\|_2 \leq 1\}$?

Self-test 8.2. Is $f(x) = \max\{|x_1|, x_2^2, e^{x_3}\}$ convex on $\mathbb{R}^3$? Is $g(x) = \min\{|x|, 1\}$ convex on $\mathbb{R}$?

Self-test 8.3. Write $\{x : \|x - v_1\| \leq \|x - v_2\|\}$ for $v_1 = (0, 0)$, $v_2 = (2, 2)$ in the form $c^\mathsf{T} x \leq d$.

Self-test 8.4. Give a convex function on $\mathbb{R}$ that is not strictly convex, and a strictly convex function whose second derivative vanishes somewhere.

Self-test 8.5. Is $\{x : \|Ax + b\|_2 \geq c^\mathsf{T} x + d\}$ convex in general? What about $\{x : \|Ax + b\|_2 \leq c^\mathsf{T} x + d\}$?

Self-test 8.6. The half-space $\{x : 2x_1 - x_2 \leq 4\}$ is contained in $\{x : \tilde{a}^\mathsf{T} x \leq \tilde{b}\}$. What can you say about $\tilde{a}$ and $\tilde{b}$?

Self-test 8.7. Distance between $\{x_1 + 2x_2 + 2x_3 = 1\}$ and $\{x_1 + 2x_2 + 2x_3 = 7\}$?

Self-test 8.8. True or false: if all sublevel sets of f are convex and f is differentiable, then f is convex.

Answers

1. Yes (intersection of a ball and a half-space). No: e_1 and $2e_1$ are in the union but $1.5e_1$ is not.
2. Yes: max of convex functions (each is convex: a norm, a convex quadratic, an exponential). No: $g(0) = 0,\ g(2) = 1,\ g(1) = 1 > \frac{1}{2}(0 + 1)$; it flattens out, a pointwise min of convex functions.
3. $c = 2(v_2 - v_1) = (4, 4)$, $d = \|v_2\|^2 - \|v_1\|^2 = 8$, i.e. $x_1 + x_2 \leq 2$. Check: $v_1 = (0, 0)$ satisfies it, $v_2 = (2, 2)$ does not.
4. $f(x) = |x|$ or any affine function (convex, not strictly); $f(x) = x^4$ (strictly convex, $f''(0) = 0$).
5. The "$\geq$" set is generally not convex (it is the complement of a convex set's interior, e.g. $\{|x| \geq 1\}$). The "$\leq$" set is convex: preimage of the second-order cone under an affine map.
6. $\tilde{a} = \lambda(2, -1)$ and $\tilde{b} \geq 4\lambda$ for some $\lambda > 0$.
7. $|1 - 7|/\|(1, 2, 2)\| = 6/3 = 2$.
8. False: $f(x) = x^3$ is smooth with sublevel sets $(-\infty, c^{1/3}]$, but $f''(x) < 0$ for $x < 0$.

9 One-page cheat sheet

Convex set	$x, y \in C,\ \lambda \in [0,1] \Rightarrow \lambda x + (1-\lambda)y \in C$
Convex function	$f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda) f(y)$; equivalently epi f convex
1st-order test	$f(y) \ge f(x) + \nabla f(x)^{\mathsf{T}}(y-x)$
2nd-order test	$\nabla^2 f \succeq 0$
Sets: preserved by	intersection (any family), affine image/preimage, sums, products, perspective
Sets: NOT preserved by	union, complement
Functions: preserved by	nonneg. sums, affine composition, pointwise max/sup, monotone convex composition, partial inf, perspective
Functions: NOT preserved by	pointwise min, products, differences, general composition
Sublevel sets	convex $f \Rightarrow L_c$ convex; converse false ($\sqrt{\lvert x\rvert}$, x^3); "all L_c convex" = quasiconvex
Hyperplane / half-space	$\{a^{\mathsf{T}}x = b\}$ / $\{a^{\mathsf{T}}x \le b\}$, normal a points to the $\ge$ side
Point–plane distance	$\lvert a^{\mathsf{T}}x - b\rvert / \lVert a\rVert_2$
Parallel planes distance	$\lvert b_1 - b_2\rvert / \lVert a\rVert_2$
Half-space containment	$\{a^{\mathsf{T}}x \le b\} \subseteq \{\tilde{a}^{\mathsf{T}}x \le \tilde{b}\} \iff \tilde{a} = \lambda a,\ \tilde{b} \ge \lambda b,\ \lambda > 0$
Closer to v_1 than v_2	$2(v_2 - v_1)^{\mathsf{T}}x \le \lVert v_2\rVert^2 - \lVert v_1\rVert^2$ (perpendicular bisector)
Separation	disjoint convex $\Rightarrow$ separating hyperplane; strict needs closed + compact
Strict-separation counterexample	$\{y \le 0\}$ vs. $\{x > 0,\ y \ge 1/x\}$: closed, disjoint, distance 0
Second-order cone set	$\{x : \lVert Ax+b\rVert_2 \le c^{\mathsf{T}}x + d\}$ convex (affine preimage of $\{\lVert u\rVert \le t\}$)

Further reading. Boyd & Vandenberghe, *Convex Optimization*, Chapter 2 (sets, §2.1–2.5) and Chapter 3 (functions, §3.1–3.4). Exercises 2.1–2.12, 2.23, 3.1–3.10 are the best drill for this material.